

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 1 / 13

REVISION HISTORY

Date	Version	Description	Author
14.03.2026	1.0	Initial STP draft created; defined document conventions and references.	Bora
15.03.2026	1.1	Completed Section 2 (Test Preparations), including hardware/software and others.	Hamza, Genidy
15.03.2026	1.2	Developed core Architectural & Functional test cases for Section 3.1.	Genidy, Emir
15.03.2026	1.3	Added Non-Functional test cases (Section 3.2).	Hamza, Bora
16.03.2026	1.4	Final team review and document polishing.	Genidy, Emir, Bora, Hamza
21.03.2026	1.5	Section 3.2 added, non-functional test procedures updated according to the black-box requirements and specifications.	Bora
25.03.2026	1.6	Updated functional tests (001-004) to strictly follow black-box GUI verification. Added T-SRS-SCMS-005 and 006. Updated TOC.	Hamza, Genidy

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 2 / 13

TABLE OF CONTENTS

1 Introduction.....	3
1.1 Document overview.....	3
1.2 References.....	3
1.3 Conventions.....	3
2 Tests preparations.....	4
2.1 Hardware preparation.....	4
2.2 Software preparation.....	4
2.3 Other test preparation.....	4
2.4 Safety, security and privacy precautions.....	4
3 Tests.....	5
3.1 Architectural & Functional Tests.....	5
3.2 Non-functional Tests.....	11

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 3 / 13

1 Introduction

1.1 Document overview

This document is the software test plan of the SCMS software development project. It contains the description of tests.

1.2 References

#	Document Identifier	Document Title
[SRS]	SCMS-SRS-1	SCMS Software Requirements Specifications
[SDP]	SCMS-SDP-v2	SCMS Software Development Plan

1.3 Conventions

Each test case will be presented in a table format as shown below:

Test ID	<ID of the test>	
Test description	<A brief description of the test>	
Verified Requirement	<ID numbers of requirements that are to be verified with this test.>	Verification method: <I,A,D,T>
Initial conditions	<The state of software before test>	
Tests inputs	<Input data from any test tool, input files name and location, if any >	
Data collection actions	<Recording and post processing of output data, if any >	
Tests outputs	<Output data files, if any>	
Assumptions and constraints	If any, may be limited access to a tool, license ...	
Expected results and criteria	<list of expected results>	<special criteria to be met by the result, if any>
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	<actions listed row by row>	<expectation after each action if any>
2	...	...

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 4 / 13

2 Tests preparations

This section contains tasks and recommendations before executing tests.

2.1 Hardware preparation

Tests will be conducted on standard workstation PCs.

- **Processor:** Intel i5 or i7 (11th gen or higher) or AMD Ryzen 5 (5000 series or higher) with at least 4 cores
- **Memory:** Minimum 16GB RAM.
- **Network:** Local Ethernet or Wi-Fi for database connectivity.
- **OS:** Windows 10 (64-bit), macOS Ventura+, or Ubuntu 22.04 LTS

2.2 Software preparation

The testing environment requires the following configuration:

- **JDK:** Main development platform for logic execution.
- **IDE:** IntelliJ IDEA or Eclipse for debugging and running automated JUnit tests.
- **Testing Framework:** JUnit for unit and component verification.
- **GUI Environment:** JavaFX framework.
- **Database:** SQL environment

2.3 Other test preparation

Compatibility for each software will be tested for every group member's personal computers.

2.4 Safety, security and privacy precautions

- **SQL Injection:** Tests involving user input must verify the use of prepared statements and parameterized queries to protect the data layer.

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 5 / 13

3 Tests

List of Tests by Test ID:

- **T-SRS-SCMS-001:** Test to check the functionality of the registration method
- **T-SRS-SCMS-002:** Test to verify event creation and UI list updates by the administrator.
- **T-SRS-SCMS-003:** Test to verify financial tracking, budget calculation, and recording of expenses.
- **T-SRS-SCMS-004:** Test to verify the membership portal, profile viewing, and role-based access restrictions.
- **T-SRS-SCMS-NF-01:** Test to verify the use of the JavaFX GUI framework for the Presentation Layer.
- **T-SRS-SCMS-NF-02:** Test to verify adherence to the 3-package architecture and Java coding conventions.
- **T-SRS-SCMS-NF-03:** Test to verify the use of parameterized queries to prevent SQL injection vulnerabilities.

3.1 Architectural & Functional Tests

Test ID	T-SRS-SCMS-001	
Test description	Verify that the registration of a new member can be fulfilled through the JavaFX GUI and successfully reflected in the system.	
Verified Requirement	SRS-SCMS-001	Verification method: D
Initial conditions	Application is running. User is logged in as an Administrator.	
Tests inputs	Member Name: "Bora", Student ID: "033620", Club: "IEEE", Role: "Member".	
Data collection actions	Visual observation of the JavaFX GUI elements and list views.	
Tests outputs	System UI success alerts and updated Member List rendering.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	Member is successfully registered. The new member's details are visibly appended to the system's Member List.	Database containing the member with Member Name: "Bora" and studentID: 033620.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 6 / 13

1	Launch SCMS program and log in as an Administrator.	Main dashboard is visible; JavaFX GUI is responsive.
2	Navigate to the "Member Registration" tab, fill in the form with the test inputs, and submit.	A success alert/dialog is displayed on the screen.
3	Navigate to the "Member List" (or "Manage Members") dashboard tab and refresh if necessary.	The newly registered member "Bora" (ID: 033620) is clearly visible in the data table on the GUI

Test ID	T-SRS-SCMS-002	
Test description	Verify that an administrator can create a new event through the JavaFX GUI, and the event list updates automatically to display the newly scheduled event.	
Verified Requirement	SRS-SCMS-002	Verification method: D
Initial conditions	Application is running; Database connection is active via the Data Layer. User is logged in as an Administrator.	
Tests inputs	Event Name: "Tech Talk", Date: "10-11-2026", Location: "Auditorium", Quota: "150".	
Data collection actions	Visual observation of the GUI event list and confirmation dialogs	
Tests outputs	System UI success alerts and rendering of the new event row in the Event Management dashboard.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	Event is successfully created. "Tech Talk" is instantly visible on the GUI event list with the specified quota and details.	Database containing the event with Event Name: "Tech Talk" and Quota: 150.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Navigate to the "Event Management" tab on the Administrator dashboard.	Event management screen and current event list are displayed.
2	Fill in the event creation form with the test inputs and submit.	A success alert/dialog is displayed on the screen.

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 7 / 13

3	Close the success alert and verify the updated "Upcoming Events" list on the GUI.	The event list refreshes automatically to include "Tech Talk" with a quota of 150.
---	---	--

Test ID	T-SRS-SCMS-003	
Test description	Verify that the application records income and expenditures through the GUI, automatically calculates the remaining budget, and displays the summary accurately.	
Verified Requirement	SRS-SCMS-003, SRS-SCMS-003.1, SRS-SCMS-003.3	Verification method: D
Initial conditions	Application is running; Database connection is active via the Data Layer. Initial budget is 1000 TL. Administrator is logged in.	
Tests inputs	Transaction Type: "Expense", Amount: "200", Description: "Pizza for meeting", Date: "12-11-2026".	
Data collection actions	Visual observation of the GUI budget dashboard and transaction history table.	
Tests outputs	Updated financial summary and transaction history row on the UI.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	Expense is successfully recorded. Remaining budget is automatically updated to 800 TL and displayed on the GUI.	Database containing the transaction with Description: "Pizza for meeting" and Amount: 200.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Navigate to the "Financial Tracking" tab.	Current budget summary (1000 TL) and transaction list are displayed.
2	Fill in the transaction form with the test inputs and submit.	A success alert is displayed. Budget summary updates to 800 TL remaining.
3	Open the database view and check the "Transactions" table.	Transaction: "Pizza for meeting" is existent with amount: 200.

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 8 / 13

Test ID	T-SRS-SCMS-004	
Test description	Verify that a logged-in member can view their own profile and upcoming events, and explicitly verify that their access is restricted from all administrator features (e.g., Financial Tracking, Manage Members).	
Verified Requirement	SRS-SCMS-004, SRS-SCMS-004.2	Verification method: D
Initial conditions	Application is running. A member account (Member Name: "Genidy") already exists. User is at the login screen.	
Tests inputs	Login credentials for Member: "Genidy".	
Data collection actions	Visual observation of rendered UI components and available navigation tabs.	
Tests outputs	Member dashboard rendering and restricted navigation menu.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	Member dashboard is loaded successfully. Profile and event data match the user. Administrator-only tabs (e.g., Financial Tracking, Member Registration) are entirely hidden or inaccessible on the UI.	Administrator-only tabs (e.g., Financial Tracking, Member Registration) are not accessible.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Launch SCMS program and log in using Member credentials.	Main member dashboard is visible; "Upcoming Events" list is displayed.
2	Navigate to the "My Profile" tab.	Member's status and basic info are displayed and match the database.
3	Inspect the main navigation menu and attempt to locate/access administrator modules (e.g., Manage Members, Event Creation, Financial Tracking).	Administrator tabs are either not visible on the GUI or clicking them results in an "Access Denied" alert dialog.

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 9 / 13

Test ID	T-SRS-SCMS-005	
Test description	Verify that the system correctly rejects invalid member registrations (e.g., Student IDs that are not exactly 8 digits or that already exist) and displays appropriate error messages.	
Verified Requirement	SRS-SCMS-001.2, SRS-SCMS-001.3	Verification method: T
Initial conditions	Application is running. User is logged in as an Administrator. The database already contains a member with Student ID "033620".	
Tests inputs	Input A (Invalid Length): Name: "Alice", ID: "12345". Input B (Duplicate): Name: "Bob", ID: "033620".	
Data collection actions	Visual observation of UI error dialogs.	
Tests outputs	System UI error alerts preventing submission.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	System blocks the registration in both scenarios. The member list is not updated, and specific error messages are shown to the user explaining why the input was rejected.	
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Navigate to the "Member Registration" tab on the Administrator dashboard.	The registration form is displayed.
2	Fill in the form using Input A (ID: "12345") and attempt to submit.	The system blocks the submission and displays a warning dialog: "Student ID must be exactly 8 numeric digits."
3	Clear the form, fill it using Input B (ID: "033620"), and attempt to submit.	The system blocks the submission and displays a warning dialog indicating that the Student ID already exists in the club.

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 10 / 13

Test ID	T-SRS-SCMS-006	
Test description	Verify that the system correctly rejects invalid event creations (e.g., scheduling in the past) and enforces attendee quotas by preventing sign-ups when an event is full.	
Verified Requirement	SRS-SCMS-002.2, SRS-SCMS-002.3	Verification method: T
Initial conditions	Application is running. User is logged in as an Administrator. An event ("Coding Bootcamp") exists with a Quota of 1 and currently has 1 member signed up.	
Tests inputs	Input A (Past Date): Event Name: "Old Event", Date: "01-01-2020", Quota: "50". Input B (Over Quota): Attempt to sign up "Hamza" to "Coding Bootcamp".	
Data collection actions	Visual observation of UI error dialogs and event lists.	
Tests outputs	System UI error alerts preventing submission.	
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.	
Expected results and criteria	System blocks both actions. The old event is not created, and the over-quota sign-up is rejected. Specific error messages are shown to the user.	
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Navigate to the "Event Management" tab on the Administrator dashboard.	The event creation form is displayed.
2	Fill in the form using Input A (Date: "01-01-2020") and attempt to submit.	The system blocks the submission and displays a warning dialog: "Cannot schedule an event in the past."
3	Log out, log back in as a Member ("Hamza"), and navigate to the "Upcoming Events" list.	The "Coding Bootcamp" event is visible.
4	Attempt to sign up for the "Coding Bootcamp" event using Input B.	The system blocks the sign-up and displays a warning dialog indicating

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 11 / 13

		the maximum attendee quota has been reached.
--	--	--

3.2 Non-functional Tests

Test ID	T-SRS-SCMS-NF-01	
Test description	Verify that the Presentation Layer is built using JavaFX and FXML files, as specified in the SDP.	
Verified Requirement	SRS-SCMS-NF-01	Verification method: I
Initial conditions	Source code is available in the IDE or GitHub repository.	
Tests inputs	Application window is launched.	
Data collection actions	Inspection of the UI properties.	
Tests outputs	Inspection results.	
Assumptions and constraints	JavaFX libraries are required for the app to launch.	
Expected results and criteria	App launches.	GUI utilizes modern JavaFX styling and FXML layouts.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Execute the application executable.	Application window is displayed successfully.
2	Resize the window and interact with the functional buttons.	UI components behave according to JavaFX layout rules (scaling/alignment).

Test ID	T-SRS-SCMS-NF-02	
Test description	Verify that the software adheres to the 3-package architecture (Presentation, Application, Data) and Oracle Java Code Conventions.	

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 12 / 13

Verified Requirement	SRS-SCMS-NF-02	Verification method: D
Initial conditions	Database is temporarily disconnected.	
Tests inputs	User interaction on the Login screen.	
Data collection actions	Observation of the error handling displayed in the GUI.	
Tests outputs	Screenshots of the handled error dialog.	
Assumptions and constraints	Layers must only communicate through stable interfaces.	
Expected results and criteria	A "Database Connection Lost" dialog is shown by the UI to the user, the app remains intact without crashing.	Database is unreachable.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Disable the local SQL database service.	Database is unreachable.
2	Attempt to log in to the SCMS application.	Login fails as database is disconnected, a descriptive error message is shown instead of application crashing.
3	Re-enable the database and attempt login again.	Login succeeds. The presentation layer functions as normal.

Test ID	T-SRS-SCMS-NF-03	
Test description	Verify that the system is secure against various kinds of SQL Injection in all user input fields.	
Verified Requirement	SRS-SCMS-NF-003	Verification method: T
Initial conditions	SCMS Login screen is active; Data Layer connection is established.	
Tests inputs	Payload 1 (Tautology): ' OR 1=1 -- Payload 2 (Auth Bypass): admin' --	

Software Test Plan of SCMS		
Doc # SCMS-STP	Version: 1.6	Page 13 / 13

	Payload 3 (Union): ' UNION SELECT NULL, username, password FROM users-- Payload 4 (Time-based Blind): '; WAITFOR DELAY '0:0:5'-- Payload 5 (XSS/SQL combo): <script>alert(1)</script>	
Data collection actions	Visual observation of GUI feedback	
Tests outputs	Screenshots of rejected login attempts and search results returned through requests.	
Assumptions and constraints	The system is required to be using PreparedStatements and Parameterized Queries as specified in the Risk Planning section.	
Expected results and criteria	Every request is treated like a literal string, login with the malicious input fails and search shows zero results.	No "SQL Syntax Error" exceptions are thrown, no unauthorized data is accessed by the attacking attempt.
Test procedure		
Step number	Operator actions	Expected result and evaluation criteria
1	Navigate to the Login screen and enter Payload 1 into the username field.	Login is denied.
2	Attempt login once more with Payload 2 .	Login is denied.
3	Enter Payload 3 into any search/filter field.	No unauthorized data is displayed.
4	Enter Payload 4 and submit.	The system responds immediately, system ignores the delay request.
5	Enter Payload 5 into "Member Name" and register.	Account creation results succesful.
6	Open "Member List" and view the entry.	Member name displays as "<script>alert(1)</script>".